

BRYAN KALEMBA

Computer Science Undergraduate | Aspiring Software Engineer & Web Developer

07458 995921 | bryankalemba95@gmail.com | github.com/BryanKalemba | linkedin.com/in/bryan-kalemba | bryankalemba.com

PROFILE

Motivated problem-solver with strong communication, teamwork, and client-facing experience, combined with hands-on technical skills across Python, C, Java, and JavaScript. Proven track record of delivering working software and data-driven solutions for real clients and simulated industry projects. Seeking a software engineering or web development internship / graduate scheme.

EDUCATION

BSc Computer Science, University of Nottingham

2025 – 2028 (Expected)

Relevant modules: Programming & Algorithms, Databases & Interfaces, Introduction to Software Engineering

TECHNICAL SKILLS

Languages: Python, C, Java, JavaScript

Frameworks & Tools: React, Node.js, Java Spring Boot, Socket.io, Yjs (CRDT), Git, Linux

Data & APIs: Pandas, REST APIs, LLM function-calling (Gemini, Ollama), Whisper speech-to-text, Formspre

WORK EXPERIENCE

Website Manager

Paula's Hair & Beauty Salon

- Managed and maintained the salon's website to gain professional web development experience, helping attract 20 new potential customers through the online booking system.
- Improved business efficiency by 25% by streamlining the appointment booking process through the website.
- Worked directly with the business owner to understand requirements and translate them into practical website features.

AI/Technology Extern

NASCAR (via Extern)

- Used Python, Pandas, and real-world APIs over an 8-week program to transform raw, messy data into actionable business intelligence.
- Developed a weighted scoring model ranking sponsors by 'visibility efficiency,' a custom metric capturing how much exposure each sponsor gets relative to their estimated investment.

Software Engineering Job Simulation

HPE (via Forage)

- Designed a RESTful web service using Java Spring Boot and the HPE GreenLake Cloud Platform in a simulated backend developer role.
- Extended the application to support uploading JSON data via HTTP POST requests, enabling dynamic data transfer to the hosted service.
- Developed and ran unit tests using JUnit and Mockito to verify application performance before deployment.

PROJECTS

Jarvis AI Assistant

- Built a voice-controlled AI assistant for Windows using LLM function-calling (Gemini / local Ollama) to translate natural speech into ~25 real system actions — app control, file management, Steam integration, persistent memory.
- Designed a provider-agnostic architecture supporting both a cloud LLM and a fully local model behind one interface, with a schema converter reconciling their differing function-calling formats.
- Diagnosed and fixed a data-corruption bug from a non-JSON-serializable object failing to round-trip through storage, adding a self-healing recovery path.

Website for Paula's Hair & Beauty Salon

- Designed and built a business website using JavaScript, integrating Formspre to power an online booking form.
- Created an original design tailored to the salon's branding and themes, ensuring a cohesive look and feel across the site.
- Site was used by 20+ potential customers, directly supporting the salon's customer acquisition and booking process.

InkSync — Real-Time Collaborative Editor

- Built a Google Docs-style real-time collaborative text editor using Yjs, a CRDT library, so concurrent edits from multiple users merge automatically with no data loss.
- Hand-implemented the Yjs sync and awareness protocol over Socket.io instead of using the standard y-websocket library, to understand the actual handshake rather than treat it as a black box.
- Diagnosed and fixed a critical sync bug where new collaborators received a blank document, tracing the handshake message-by-message to the root cause.